

Notes: Berry, Levinsohn, and Pakes (Econometrica, 1995)

Automobile Prices in Market Equilibrium

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1 Big picture

- **Question.** How can researchers estimate demand and supply in a differentiated-products oligopoly when they observe product-level prices, quantities, and characteristics, but not individual purchases matched to individual characteristics?
- **Setting.** The paper studies the U.S. new automobile market from 1971 to 1990. The empirical unit is a model-year, such as the 1990 Mazda 323 or the 1990 BMW 735i.
- **Main contribution.** The paper develops what became the standard BLP framework: a random-coefficients discrete-choice demand system, a multiproduct Bertrand-Nash pricing equation, an inversion from observed market shares to mean utilities, and GMM estimation using product-level aggregate data.
- **Core empirical challenge 1: price endogeneity.** Cars with high unobserved quality, style, reputation, or reliability tend to have both higher demand and higher prices. A regression of market share on price and characteristics therefore makes demand look too price-inelastic.
- **Core empirical challenge 2: substitution patterns.** Simple logit demand implies that substitution depends mainly on market shares, not on similarity in product characteristics. It can predict that a BMW price increase shifts demand toward a Yugo and a Mercedes in similar ways if their market shares are similar.
- **Demand-side solution.** Consumers have heterogeneous tastes over observed characteristics. The utility from a car depends on price, characteristics, unobserved product quality, income, and random taste shocks. Aggregating these choices gives product-level market shares.
- **Supply-side solution.** Firms are multiproduct oligopolists that set prices in a Bertrand-Nash equilibrium. The first-order conditions imply model-specific markups. These markups, combined with observed prices, back out marginal costs.
- **Identification logic.** Product characteristics and cost shifters are assumed mean independent of unobserved demand and cost shocks. Prices are instrumented with functions of the characteristics of own-firm and rival products, which affect markups and equilibrium prices but are excluded from a product's own utility shock.
- **Algorithmic innovation.** For a trial parameter vector, the paper solves for the vector of mean utilities that makes predicted market shares equal observed market shares. The key computational step is the contraction mapping

$$\delta_j^{new} = \delta_j^{old} + \log(s_j^{obs}) - \log(s_j^{pred}).$$

- **Main logit result.** Ordinary logit estimates imply that **1,494 of 2,217** model-year demands are inelastic. This is inconsistent with static profit maximization by firms selling differentiated products.
- **Main IV-logit result.** Instrumenting price more than doubles the absolute price coefficient, and the number of inelastic demands falls to **22**. This shows that unobserved product quality and price endogeneity are economically important.

- **Main BLP result.** The full random-coefficients model produces elastic demand for all products, plausible own- and cross-price elasticities, and markups that rise with price and product differentiation.
- **Economic takeaway.** The paper provides a practical structural method for estimating demand, marginal costs, markups, and counterfactual equilibria in differentiated-products markets using only aggregate product-level data.

2 Data and measurement (market, products, and key objects)

2.1 Why automobiles are the right laboratory

- The automobile market has many horizontally and vertically differentiated products.
- Consumers differ strongly in tastes for price, size, power, fuel economy, and amenities.
- Firms sell multiple products, so the correct pricing problem is a multiproduct oligopoly problem.
- Product-level data are relatively rich and available for a long period.
- The market is economically important and relevant for trade policy, merger policy, fuel-economy regulation, and price-index construction.

Interpretation. Cars are not interchangeable homogeneous goods. Any credible empirical model must allow a compact economy car, a family sedan, and a luxury sedan to have very different substitution relationships.

2.2 Sample construction

- The paper uses essentially all new car models marketed in the United States from **1971 to 1990**.
- The sample excludes exotic models with extremely small market shares, such as Ferrari and Rolls Royce.
- The total sample contains **2,217 model-year observations**.
- Using a model-linking rule based on model name and changes in horsepower, width, length, and wheelbase, these observations correspond to **997 distinct models**.
- The panel is unbalanced because models enter and exit over time.

Interpretation. The panel structure is useful because it provides variation over both products and time, while also allowing unobserved characteristics of the same model to be correlated across adjacent years.

2.3 Product characteristics

- Main product data come from annual issues of the *Automotive News Market Data Book*.
- Observed characteristics include cylinders, doors, weight, engine displacement, horsepower, length, width, wheelbase, EPA miles per gallon, and indicators for front-wheel drive, automatic transmission, power steering, and air conditioning.
- The baseline demand specification uses a parsimonious set of characteristics: horsepower per weight, air conditioning, miles per dollar of gasoline, size, and a constant.
- Size is measured as length times width.
- Miles per dollar is MPG divided by the real price of gasoline.

Interpretation. The paper keeps the baseline characteristic space small because each random coefficient adds a difficult nonlinear parameter. The chosen variables proxy for acceleration, luxury, fuel economy, and size or safety.

2.4 Prices, quantities, and market size

- Price is the list retail price of the base model, measured in thousands of 1983 dollars.
- Quantity is U.S. sales by nameplate, measured in thousands of units.
- The market size M is the number of U.S. households.
- The outside good is the option of not buying a new car in that year.
- The outside share equals one minus the total new-car purchase share.

Interpretation. The outside good is crucial because it lets total automobile demand respond to prices. Without it, the model would only reallocate a fixed number of purchases across models.

2.5 Consumer heterogeneity and income

- The paper does not observe which households buy which cars.
- Instead, it integrates individual choices over a distribution of consumer characteristics.
- Income is taken from the March Current Population Survey and modeled as lognormal within each year.
- Other random tastes are assumed normal and independent of income.

Interpretation. The model uses aggregate product-level data but still incorporates micro information about the income distribution. This is one reason the model can distinguish demand for cheap cars from demand for expensive cars.

2.6 Cost shifters and firm ownership

- The supply side uses observed cost characteristics w_j .
- The baseline cost function includes horsepower per weight, air conditioning, MPG, size, a time trend, and a constant.
- Continuous cost shifters enter the marginal-cost equation in logs.
- Firm ownership matters because markups depend on which models are jointly priced by the same multiproduct firm.
- Different brands owned by the same parent company are treated as one firm. For example, Buick, Cadillac, Chevrolet, Oldsmobile, and Pontiac are all part of General Motors.

Interpretation. A firm's incentive to raise the price of one model depends on whether lost buyers switch to a rival's product or to another model owned by the same firm.

2.7 Descriptive facts

- The number of models rises from **92** in 1971 to a high of **150** in 1988.
- Sales per model generally decline over the sample period.
- The sales-weighted real list price rises sharply in the 1980s, from roughly **7.7–8.3** thousand dollars early in the sample to above **10** thousand dollars by 1989–1990.
- Domestic firms' sales share falls from a high of **93.2%** in 1973 to **68.2%** in 1990.
- Japanese firms' share rises from **4.0%** in 1973 to **27.6%** in 1990.
- Average fuel efficiency improves substantially after the oil-price shocks.
- Air conditioning becomes much more common as standard equipment, rising from zero at the start of the sample to **30.8%** by 1990.

Interpretation. The sample contains major changes in product variety, foreign competition, fuel economy, and amenities. These changes provide useful variation for estimating demand and supply.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Individual utility and market demand

Consumer i 's utility from product j is written generically as

$$U_{ij} = U(\zeta_i, p_j, x_j, \xi_j; \theta), \tag{1}$$

where ζ_i denotes consumer characteristics, p_j is price, x_j is observed product characteristics, ξ_j is unobserved product quality, and θ is the vector of parameters.

Consumer i chooses product j if

$$U_{ij} \geq U_{ir} \quad \text{for all } r = 0, 1, \dots, J, \quad (2)$$

where $j = 0$ is the outside good. The set of consumer types choosing j is

$$A_j = \{\zeta : U(\zeta, p_j, x_j, \xi_j; \theta) \geq U(\zeta, p_r, x_r, \xi_r; \theta), \forall r\}. \quad (3)$$

The market share of product j is

$$s_j(p, x, \xi; \theta) = \int_{A_j} dP_\theta(\zeta). \quad (4)$$

Interpretation. Demand is obtained by aggregating individual discrete choices. Product j 's share is the mass of consumers for whom j gives the highest utility.

3.2 Simple logit benchmark

The simple logit utility is

$$u_{ij} = x_j\beta - \alpha p_j + \xi_j + \varepsilon_{ij}, \quad (5)$$

where ε_{ij} is i.i.d. extreme value. Define mean utility

$$\delta_j = x_j\beta - \alpha p_j + \xi_j. \quad (6)$$

The logit market share is

$$s_j = \frac{\exp(\delta_j)}{1 + \sum_{k=1}^J \exp(\delta_k)}, \quad (7)$$

and the inversion is analytic:

$$\delta_j = \log(s_j) - \log(s_0). \quad (8)$$

Thus the logit demand error is

$$\xi_j = \log(s_j) - \log(s_0) - x_j\beta + \alpha p_j. \quad (9)$$

Interpretation. Logit is computationally convenient because observed shares directly reveal mean utility. The problem is that it imposes unrealistic substitution patterns: consumers who leave one product tend to substitute toward products with high overall market shares rather than toward similar products.

3.3 Random-coefficients demand

The random-coefficients specification allows consumer tastes for characteristics to vary:

$$u_{ij} = x_j\beta - \alpha p_j + \xi_j + \sum_k \sigma_k x_{jk} v_{ik} + \varepsilon_{ij}. \quad (10)$$

Here v_{ik} is a standardized taste shock and σ_k measures heterogeneity in tastes for characteristic k .

Interpretation. If some consumers especially like large cars, they attach high utility to all large cars. A price increase for one large car therefore shifts more demand toward other large cars. This solves the main substitution-pattern defect of simple logit.

3.4 Cobb-Douglas specification with income

The empirical model uses a Cobb-Douglas form in expenditure on other goods. For inside goods,

$$u_{ijt} = \alpha \log(y_{it} - p_{jt}) + x_{jt}\beta + \xi_{jt} + \sum_k \sigma_k x_{jkt} v_{ikt} + \varepsilon_{ijt}. \quad (11)$$

For the outside good,

$$u_{i0t} = \alpha \log(y_{it}) + \sigma_0 v_{i0t} + \varepsilon_{i0t}. \quad (12)$$

Income is represented as

$$y_{it} = \exp(m_t + \sigma_y v_{iyt}), \quad (13)$$

where m_t and σ_y come from the CPS income distribution.

Interpretation. The marginal utility of price depends on income. A \$1,000 price increase is more costly for low-income households than for high-income households, so price sensitivity varies across consumers.

3.5 Market share simulation

Conditional on consumer heterogeneity v_i , the extreme-value shocks can be integrated out analytically:

$$f_j(v_i, \delta, p, x; \theta) = \frac{\exp\{\delta_j + \mu_{ij}\}}{1 + \sum_{k=1}^J \exp\{\delta_k + \mu_{ik}\}}, \quad (14)$$

where μ_{ij} is the consumer-specific deviation from mean utility. The unconditional share is

$$s_j(p, x, \delta; \theta) = \int f_j(v, \delta, p, x; \theta) dP(v). \quad (15)$$

The simulated share is

$$\hat{s}_j(p, x, \delta; \theta) = \frac{1}{N_s} \sum_{i=1}^{N_s} f_j(v_i, \delta, p, x; \theta). \quad (16)$$

Interpretation. The model does not simulate the idiosyncratic logit shock. It only simulates the lower-dimensional random tastes and income shocks, which greatly improves computational

precision.

3.6 BLP inversion and contraction mapping

For a given nonlinear parameter vector, the model solves for the mean utilities δ that match observed shares:

$$s_j^{obs} = \widehat{s}_j(p, x, \delta; \theta). \quad (17)$$

The fixed-point iteration is

$$\delta_j^{(r+1)} = \delta_j^{(r)} + \log(s_j^{obs}) - \log(\widehat{s}_j(p, x, \delta^{(r)}; \theta)). \quad (18)$$

Then

$$\xi_j(\theta) = \delta_j(\theta) - x_j\beta. \quad (19)$$

Interpretation. This is the central computational insight of the paper. For every proposed nonlinear parameter vector, the algorithm recovers the unobserved product quality vector that rationalizes the observed market shares.

3.7 Marginal cost

The marginal cost of product j is assumed log-linear:

$$\log(mc_j) = w_j\gamma + \omega_j, \quad (20)$$

where w_j is observed cost characteristics and ω_j is an unobserved cost shock.

Interpretation. Product characteristics such as horsepower, size, and amenities can affect cost. The cost shock captures productivity, engineering, and production-cost components not observed by the econometrician.

3.8 Multiproduct Bertrand pricing

Let firm f own the set of products $\mathcal{F}_f$. Its profits are

$$\Pi_f = \sum_{j \in \mathcal{F}_f} (p_j - mc_j) M s_j(p, x, \xi; \theta). \quad (21)$$

The first-order condition for product $j \in \mathcal{F}_f$ is

$$s_j(p, x, \xi; \theta) + \sum_{r \in \mathcal{F}_f} (p_r - mc_r) \frac{\partial s_r(p, x, \xi; \theta)}{\partial p_j} = 0. \quad (22)$$

Define the ownership-adjusted derivative matrix Δ by

$$\Delta_{jr} = -\frac{\partial s_r}{\partial p_j} \mathbf{1}\{j \text{ and } r \text{ are owned by the same firm}\}. \quad (23)$$

Then the vector of markups is

$$b(p, x, \xi; \theta) = \Delta^{-1} s(p, x, \xi; \theta), \quad (24)$$

and prices satisfy

$$p = mc + b(p, x, \xi; \theta). \quad (25)$$

Interpretation. Markups are high when consumers are less price-sensitive and when a firm's own other products capture many of the consumers who switch away after a price increase.

3.9 Pricing equation used in estimation

Using $mc_j = p_j - b_j$, the supply equation becomes

$$\log(p_j - b_j(p, x, \xi; \theta)) = w_j \gamma + \omega_j. \quad (26)$$

Interpretation. Once the demand parameters imply a markup, the difference between price and markup is marginal cost. The residual from the cost equation is the unobserved cost shock.

3.10 Moment conditions and GMM

The key identifying restrictions are

$$E[\xi_j | z] = 0, \quad E[\omega_j | z] = 0, \quad (27)$$

where z contains observed product characteristics and cost shifters, but not price or quantity.

Using instruments Z_j , the moment conditions are

$$E[Z_j' \xi_j(\theta)] = 0, \quad E[Z_j' \omega_j(\theta)] = 0. \quad (28)$$

The estimator chooses parameters to minimize a quadratic form in the sample moments.

Interpretation. Prices and quantities are endogenous equilibrium outcomes, so they are not valid instruments. Characteristics of products in the market are used because they affect equilibrium markups and prices through competition.

3.11 Concentrating out linear parameters

The nonlinear search is over the parameters that govern consumer heterogeneity and price-income sensitivity. Conditional on these nonlinear parameters, the linear demand and cost parameters can be estimated by linear IV/GMM steps.

Interpretation. This reduces the dimension of the difficult nonlinear optimization problem. The computational burden is concentrated in solving the share inversion and computing markups.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies demand

- Demand is identified from how market shares vary with prices and characteristics across models and years.
- The key challenge is that price is correlated with unobserved quality ξ_j .
- The model uses instruments based on observed characteristics of all products in the market.
- The characteristics of other products affect equilibrium prices because they determine the degree of competition faced by product j .
- Conditional on product j 's own characteristics, rival products' characteristics are excluded from j 's own demand shock.

Interpretation. The identification is not a simple exclusion from demand levels. Rival characteristics enter the demand system through substitution and competition, but they do not enter product j 's unobserved quality shock.

4.2 Why price endogeneity matters

- If high-quality unobserved characteristics raise both demand and price, OLS underestimates price sensitivity.
- This is exactly what the paper finds: the OLS logit price coefficient is too small in absolute value.
- Instrumenting price causes the absolute price coefficient to rise substantially.
- The number of inelastic demands falls from 1,494 under OLS logit to 22 under IV logit.

Interpretation. The evidence strongly supports the classic simultaneity story: high unobserved quality makes expensive cars sell well, so naive regressions confuse quality with low price sensitivity.

4.3 Why simple logit is not enough

- IV logit corrects price endogeneity but still imposes logit's substitution structure.
- In logit, cross-price effects depend heavily on market shares and not enough on product similarity.
- The IV-logit estimates imply nearly uniform markups across very different products.
- This is economically implausible because luxury cars, compact cars, and family sedans face different competitive environments.

Interpretation. There are two separate problems: price endogeneity and unrealistic substitution. Instruments fix the first, but random coefficients are needed for the second.

4.4 What identifies random coefficients

- Random coefficients are identified by how substitution patterns vary with product characteristics and prices.
- A large standard deviation of tastes for fuel economy means that buyers of fuel-efficient cars are especially likely to substitute toward other fuel-efficient cars.
- A large standard deviation of tastes for size means that consumers sort into different regions of product space.
- The supply-side pricing equations also help because markups reveal information about the slopes of demand.

Interpretation. Mean tastes help explain average market shares. Taste dispersion helps explain which products are close substitutes and therefore how firms can mark up prices.

4.5 Supply-side identification

- Given demand parameters and ownership, the pricing first-order conditions imply markups.
- Marginal costs are recovered as $p_j - b_j$.
- The cost equation relates these recovered marginal costs to observed cost shifters.
- The cost shocks ω_j are assumed mean independent of the same class of observed characteristics and cost shifters.

Interpretation. The supply side turns estimated demand elasticities into marginal costs and markups. This is why demand estimation and oligopoly pricing are tightly linked in the paper.

4.6 Main maintained assumptions

- Product characteristics and cost shifters are treated as exogenous with respect to unobserved demand and cost shocks.
- Firms are assumed to set prices according to static multiproduct Bertrand-Nash behavior.
- Marginal cost is log-linear in observed characteristics in the baseline specification.
- The income distribution is lognormal and estimated from the CPS.
- Idiosyncratic utility shocks are extreme value, which gives closed-form conditional choice probabilities.

Interpretation. The model is structural and powerful, but it is not assumption-free. The credibility of counterfactuals depends on the plausibility of these behavioral and exogeneity assumptions.

4.7 Important limitations

- Prices are list prices, not transaction prices.
- Fleet sales are not separately observed.
- Model-level production data are incomplete, especially for foreign producers, so output is not in the baseline cost function.
- Product characteristics are treated as fixed when solving short-run pricing equilibria.
- The outside good is modeled in a simple way, even though many consumers' outside option is keeping or buying a used car.

Interpretation. The paper is most credible for estimating short-run demand, markups, and marginal costs conditional on the observed product set. It is less complete as a full dynamic model of product design, used-car markets, and consumer replacement timing.

5 Tables and results

5.1 Tables I and II: descriptive statistics and characteristic ranges

TABLE I
DESCRIPTIVE STATISTICS

Year	No. of Models	Quantity	Price	Domestic	Japan	European	HP/Wt	Size	Air	MPG	MP\$
1971	92	86,892	7,868	0.866	0.057	0.077	0.490	1.496	0.000	1.662	1.850
1972	89	91,763	7,979	0.892	0.042	0.066	0.391	1.510	0.014	1.619	1.875
1973	86	92,785	7,535	0.932	0.040	0.028	0.364	1.529	0.022	1.589	1.819
1974	72	105,119	7,506	0.887	0.050	0.064	0.347	1.510	0.026	1.568	1.453
1975	93	84,775	7,821	0.853	0.083	0.064	0.337	1.479	0.054	1.584	1.503
1976	99	93,382	7,787	0.876	0.081	0.043	0.338	1.508	0.059	1.759	1.696
1977	95	97,727	7,651	0.837	0.112	0.051	0.340	1.467	0.032	1.947	1.835
1978	95	99,444	7,645	0.855	0.107	0.039	0.346	1.405	0.034	1.982	1.929
1979	102	82,742	7,599	0.803	0.158	0.038	0.348	1.343	0.047	2.061	1.657
1980	103	71,567	7,718	0.773	0.191	0.036	0.350	1.296	0.078	2.215	1.466
1981	116	62,030	8,349	0.741	0.213	0.046	0.349	1.286	0.094	2.363	1.559
1982	110	61,893	8,831	0.714	0.235	0.051	0.347	1.277	0.134	2.440	1.817
1983	115	67,878	8,821	0.734	0.215	0.051	0.351	1.276	0.126	2.601	2.087
1984	113	85,933	8,870	0.783	0.179	0.038	0.361	1.293	0.129	2.469	2.117
1985	136	78,143	8,938	0.761	0.191	0.048	0.372	1.265	0.140	2.261	2.024
1986	130	83,756	9,382	0.733	0.216	0.050	0.379	1.249	0.176	2.416	2.856
1987	143	67,667	9,965	0.702	0.245	0.052	0.395	1.246	0.229	2.327	2.789
1988	150	67,078	10,069	0.717	0.237	0.045	0.396	1.251	0.237	2.334	2.919
1989	147	62,914	10,321	0.690	0.261	0.049	0.406	1.259	0.289	2.310	2.806
1990	131	66,377	10,337	0.682	0.276	0.043	0.419	1.270	0.308	2.270	2.852
All	2217	78,804	8,604	0.790	0.161	0.049	0.372	1.357	0.116	2.099	2.086

Note: The entry in each cell of the last nine columns is the sales weighted mean.

TABLE II
THE RANGE OF CONTINUOUS DEMAND CHARACTERISTICS
(AND ASSOCIATED MODELS)

Variable	Percentile				
	0	25	50	75	100
Price	90 Yugo 3,393	79 Mercury Capri 6,711	87 Buick Skylark 8,728	71 Ford T-Bird 13,074	89 Porsche 911 Cabriolet 68,597
Sales	73 Toyota 1600CR 1049	72 Porsche Rdstr 15,479	77 Plym. Arrow 47,345	82 Buick LeSabre 109,002	71 Chevy Impala 577,313
HP/Wt.	85 Plym. Gran Fury 0.170	85 Suburu DH 0.337	86 Plym. Caravelle 0.375	89 Toyota Camry 0.428	89 Porsche 911 Turbo 0.948
Size	73 Honda Civic 0.756	77 Renault GTL 1.131	89 Hyundai Sonata 1.270	81 Pontiac F-Bird 1.453	73 Imperial 1.888
MP\$	74 Cad. Eldorado 8.46	78 Buick Skyhawk 15.57	82 Mazda 626 20.10	84 Pontiac 2000 24.86	89 Geo Metro 64.37
MPG	74 Cad. Eldorado 9	79 BMW 528i 17	81 Dodge Challenger 20	75 Suburu DL 25	89 Geo Metro 53

Notes: The top entry for each cell gives the model name and the number directly below it gives the value of the variable for this model.

Read:

- Table I reports sales-weighted means by year for models, quantity, price, origin shares, horsepower per weight, size, air conditioning, MPG, and miles per dollar.
- The average number of products rises over time, while sales per model tend to fall.
- Real prices rise sharply in the 1980s.
- Domestic market share declines, while Japanese market share rises strongly.
- Table II shows the range of key continuous variables. For example, the 1990 Yugo is the least expensive car in the sample, while the Porsche 911 Turbo Cabriolet is at the high end of the price distribution.

Economic message. The automobile market changes substantially over the sample. A model that can handle changing product variety and changing characteristics is necessary.

5.2 Table III: logit demand, IV logit demand, and marginal-cost pricing

Read:

- OLS logit estimates a price coefficient of **-0.089**.
- IV logit estimates a price coefficient of **-0.216**, more than twice as large in absolute value.
- OLS logit implies **1,494** inelastic demands out of 2,217.
- IV logit implies only **22** inelastic demands.
- In OLS logit, some characteristic signs are implausible; in IV logit, the main characteristics enter utility positively.

Economic message. Correcting price endogeneity is essential. High unobserved quality raises both price and demand, so OLS makes consumers appear much less price-sensitive than they are.

TABLE III
RESULTS WITH LOGIT DEMAND AND MARGINAL COST PRICING
(2217 OBSERVATIONS)

Variable	OLS Logit Demand	IV Logit Demand	OLS ln (price) on w
Constant	-10.068 (0.253)	-9.273 (0.493)	1.882 (0.119)
<i>HP/Weight*</i>	-0.121 (0.277)	1.965 (0.909)	0.520 (0.035)
<i>Air</i>	-0.035 (0.073)	1.289 (0.248)	0.680 (0.019)
<i>MP\$</i>	0.263 (0.043)	0.052 (0.086)	—
<i>MPG*</i>	—	—	-0.471 (0.049)
<i>Size*</i>	2.341 (0.125)	2.355 (0.247)	0.125 (0.063)
<i>Trend</i>	—	—	0.013 (0.002)
<i>Price</i>	-0.089 (0.004)	-0.216 (0.123)	—
<i>No. Inelastic Demands</i> (+ / - 2 s.e.'s)	1494 (1429-1617)	22 (7-101)	<i>n.a.</i>
R^2	0.387	<i>n.a.</i>	.656

Notes: The standard errors are reported in parentheses.
*The continuous product characteristics—hp/weight, size, and fuel efficiency (*MP\$* or *MPG*)—enter the demand equations in levels, but enter the column 3 price regression in natural logs.

TABLE IV
ESTIMATED PARAMETERS OF THE DEMAND AND PRICING EQUATIONS:
BLP SPECIFICATION, 2217 OBSERVATIONS

Demand Side Parameters	Variable	Parameter Estimate	Standard Error	Parameter Estimate	Standard Error
Means ($\bar{\beta}$'s)	<i>Constant</i>	-7.061	0.941	-7.304	0.746
	<i>HP/Weight</i>	2.883	2.019	2.185	0.896
	<i>Air</i>	1.521	0.891	0.579	0.632
	<i>MP\$</i>	-0.122	0.320	-0.049	0.164
	<i>Size</i>	3.460	0.610	2.604	0.285
Std. Deviations (σ_β 's)	<i>Constant</i>	3.612	1.485	2.009	1.017
	<i>HP/Weight</i>	4.628	1.885	1.586	1.186
	<i>Air</i>	1.818	1.695	1.215	1.149
	<i>MP\$</i>	1.050	0.272	0.670	0.168
	<i>Size</i>	2.056	0.585	1.510	0.297
Term on Price (α)	ln ($y - p$)	43.501	6.427	23.710	4.079
Cost Side Parameters					
	<i>Constant</i>	0.952	0.194	0.726	0.285
	ln (<i>HP/Weight</i>)	0.477	0.056	0.313	0.071
	<i>Air</i>	0.619	0.038	0.290	0.052
	ln (<i>MPG</i>)	-0.415	0.055	0.293	0.091
	ln (<i>Size</i>)	-0.046	0.081	1.499	0.139
	<i>Trend</i>	0.019	0.002	0.026	0.004
	ln (q)			-0.387	0.029

5.3 Table IV: full BLP demand and pricing estimates

Read:

- The full model estimates both means and standard deviations of tastes for characteristics.

- In the constant-returns baseline, the coefficient on $\log(y - p)$ is **43.501** with standard error **6.427**.
- The mean utility coefficients for air conditioning and size are positive; the mean coefficient for horsepower per weight is positive but less precise.
- The standard deviations for horsepower per weight, miles per dollar, and size are large and precisely estimated.
- On the cost side, horsepower per weight and air conditioning increase marginal cost. The baseline MPG coefficient is negative, which the authors interpret cautiously because output is omitted from the baseline cost function.
- When $\log(q)$ is added to the cost function, the estimates imply strong decreasing returns, but the authors are cautious because model-level output is not well measured for foreign cars.

Economic message. The data support substantial heterogeneity in consumer tastes. This heterogeneity is central for generating realistic substitution patterns and markups.

5.4 Table V: attribute and price elasticities

Read:

- The table reports elasticities for a sample of 1990 models.
- Compact cars such as the Mazda 323, Nissan Sentra, and Ford Escort have high absolute price elasticities, around **6.0–6.5**.
- More expensive luxury cars such as the Lexus LS400 and BMW 735i have lower absolute price elasticities, around **3.1–3.5**.
- Demand for fuel-efficient cars is much more sensitive to miles per dollar than demand for luxury cars.
- Demand elasticities with respect to size and horsepower per weight differ systematically across models.

Economic message. The model produces intuitive heterogeneity: economy-car buyers care more about fuel economy, while luxury-car buyers are less sensitive to fuel costs and somewhat less price elastic.

5.5 Table VI: own- and cross-price semi-elasticities

Read:

- The table reports the percentage change in the row car's market share caused by a \$1,000 price increase of the column car.
- Cross-price effects are largest between similar cars.

TABLE V
A SAMPLE FROM 1990 OF ESTIMATED DEMAND ELASTICITIES
WITH RESPECT TO ATTRIBUTES AND PRICE
(BASED ON TABLE IV (CRTS) ESTIMATES)

Model	Value of Attribute/Price Elasticity of demand with respect to:				Price
	HP/Weight	Air	MP \$	Size	
Mazda323	0.366	0.000	3.645	1.075	5.049
	0.458	0.000	1.010	1.338	6.358
Sentra	0.391	0.000	3.645	1.092	5.661
	0.440	0.000	0.905	1.194	6.528
Escort	0.401	0.000	4.022	1.116	5.663
	0.449	0.000	1.132	1.176	6.031
Cavalier	0.385	0.000	3.142	1.179	5.797
	0.423	0.000	0.524	1.360	6.433
Accord	0.457	0.000	3.016	1.255	9.292
	0.282	0.000	0.126	0.873	4.798
Taurus	0.304	0.000	2.262	1.334	9.671
	0.180	0.000	-0.139	1.304	4.220
Century	0.387	1.000	2.890	1.312	10.138
	0.326	0.701	0.077	1.123	6.755
Maxima	0.518	1.000	2.513	1.300	13.695
	0.322	0.396	-0.136	0.932	4.845
Legend	0.510	1.000	2.388	1.292	18.944
	0.167	0.237	-0.070	0.596	4.134
TownCar	0.373	1.000	2.136	1.720	21.412
	0.089	0.211	-0.122	0.883	4.320
Seville	0.517	1.000	2.011	1.374	24.353
	0.092	0.116	-0.053	0.416	3.973
LS400	0.665	1.000	2.262	1.410	27.544
	0.073	0.037	-0.007	0.149	3.085
BMW 735i	0.542	1.000	1.885	1.403	37.490
	0.061	0.011	-0.016	0.174	3.515

Notes: The value of the attribute or, in the case of the last column, price, is the top number and the number below it is the elasticity of demand with respect to the attribute (or, in the last column, price.)

TABLE VI
A SAMPLE FROM 1990 OF ESTIMATED OWN- AND CROSS-PRICE SEMI-ELASTICITIES
BASED ON TABLE IV (CRTS) ESTIMATES

	Mazda 323	Nissan Sentra	Ford Escort	Chevy Cavalier	Honda Accord	Ford Taurus	Buick Century	Nissan Maxima	Acura Legend	Lincoln Town Car	Cadillac Seville	Lincoln LS400	BMW 735i
323	-125.933	1.518	8.954	9.680	2.185	0.852	0.485	0.056	0.009	0.012	0.002	0.002	0.000
Sentra	0.705	-115.319	8.024	8.435	2.473	0.909	0.516	0.093	0.015	0.019	0.003	0.003	0.000
Escort	0.713	1.375	-106.497	7.570	2.298	0.708	0.445	0.082	0.015	0.015	0.003	0.003	0.000
Cavalier	0.754	1.414	7.406	-110.972	2.291	1.083	0.666	0.087	0.015	0.023	0.004	0.003	0.000
Accord	0.120	0.293	1.590	1.621	-51.637	1.532	0.463	0.310	0.095	0.169	0.034	0.030	0.005
Taurus	0.063	0.144	0.653	1.020	2.041	-43.854	0.335	0.245	0.091	0.291	0.045	0.024	0.006
Century	0.099	0.228	1.146	1.700	1.722	0.937	-66.635	0.773	0.152	0.278	0.039	0.029	0.005
Maxima	0.013	0.046	0.236	0.256	1.293	0.768	0.866	-35.378	0.271	0.579	0.116	0.115	0.020
Legend	0.004	0.014	0.083	0.084	0.736	0.532	0.318	0.506	-21.820	0.775	0.183	0.210	0.043
TownCar	0.002	0.006	0.029	0.046	0.475	0.614	0.210	0.389	0.280	-20.175	0.226	0.168	0.048
Seville	0.001	0.005	0.028	0.035	0.425	0.420	0.131	0.351	0.296	1.011	-16.313	0.263	0.068
LS400	0.001	0.003	0.018	0.019	0.302	0.185	0.079	0.280	0.274	0.606	0.212	-11.199	0.086
735i	0.000	0.002	0.009	0.012	0.203	0.176	0.050	0.190	0.223	0.685	0.215	0.336	-9.376

Note: Cell entries ϵ_{ij} , where i indexes row and j column, give the percentage change in market share of i with a 1000 change in the price of j .

- A price increase for a Mazda 323 shifts much more demand to a Nissan Sentra or Ford Escort than to a Lincoln Town Car, Cadillac Seville, Lexus LS400, or BMW 735i.
- The Lexus is a close substitute for the BMW 735i; the Cadillac is a close substitute for the Lincoln; and the Honda Accord is a close substitute for the Ford Taurus.

Economic message. This is the core payoff from random coefficients. Substitution occurs within economically sensible regions of product space rather than mechanically toward products with high market share.

5.6 Table VII: substitution to the outside good

Read:

- Under simple logit, roughly **90%** of consumers who leave any model after a price increase substitute to the outside good.
- Under the BLP model, outside-good substitution is much lower and varies across models.
- For the sample of 1990 models, the BLP outside substitution share is about **27%** for the Mazda 323, **25%** for the Ford Taurus, **17%** for the Cadillac Seville, and about **10%** for the Lexus LS400 and BMW 735i.

Economic message. Simple logit pushes too much substitution out of the new-car market. BLP keeps more substitution within relevant car categories, though the authors still view the outside-good treatment as an area for improvement.

TABLE VII
SUBSTITUTION TO THE OUTSIDE GOOD

Model	Given a price increase, the percentage who substitute to the outside good (as a percentage of all who substitute away.)	
	Logit	BLP
Mazda 323	90.870	27.123
Nissan Sentra	90.843	26.133
Ford Escort	90.592	27.996
Chevy Cavalier	90.585	26.389
Honda Accord	90.458	21.839
Ford Taurus	90.566	25.214
Buick Century	90.777	25.402
Nissan Maxima	90.790	21.738
Acura Legend	90.838	20.786
Lincoln Town Car	90.739	20.309
Cadillac Seville	90.860	16.734
Lexus LS400	90.851	10.090
BMW 735i	90.883	10.101

TABLE VIII
A SAMPLE FROM 1990 OF ESTIMATED PRICE-MARGINAL-COST MARKUPS AND VARIABLE PROFITS: BASED ON TABLE 6 (CRTS) ESTIMATES

	Price	Markup Over MC ($p - MC$)	Variable Profits (in \$'000's) $q * (p - MC)$
Mazda 323	\$5,049	\$ 801	\$18,407
Nissan Sentra	\$5,661	\$ 880	\$43,554
Ford Escort	\$5,663	\$1,077	\$311,068
Chevy Cavalier	\$5,797	\$1,302	\$384,263
Honda Accord	\$9,292	\$1,992	\$830,842
Ford Taurus	\$9,671	\$2,577	\$807,212
Buick Century	\$10,138	\$2,420	\$271,446
Nissan Maxima	\$13,695	\$2,881	\$288,291
Acura Legend	\$18,944	\$4,671	\$250,695
Lincoln Town Car	\$21,412	\$5,596	\$832,082
Cadillac Seville	\$24,353	\$7,500	\$249,195
Lexus LS400	\$27,544	\$9,030	\$371,123
BMW 735i	\$37,490	\$10,975	\$114,802

5.7 Table VIII: markups and variable profits

Read:

TABLE IX
RESULTS FROM SOME ALTERNATIVE SPECIFICATIONS:
PRICE-MARGINAL-COST MARKUPS

	Base Case (reported in Table VIII)	Include ln(<i>q</i>) in cost function	Use <i>AT</i> instead of <i>AIR</i>	Weight and <i>HP</i> instead of <i>HP/Wt</i>	Include interaction terms in cost function	Use 3 region dummies and add <i>Reliability</i>	Add weight and include interactions in cost function
Mazda 323	\$ 801	\$1,616	\$1,012	\$1,073	\$ 828	\$1,125	\$1,389
Nissan Sentra	\$ 880	\$1,769	\$1,153	\$1,271	\$ 912	\$1,308	\$1,487
Ford Escort	\$1,077	\$2,043	\$1,326	\$1,470	\$1,111	\$2,094	\$1,690
Chevy Cavalier	\$1,302	\$2,490	\$1,729	\$1,655	\$1,329	\$2,593	\$2,020
Honda Accord	\$1,992	\$3,059	\$2,629	\$2,703	\$2,059	\$3,839	\$2,327
Ford Taurus	\$2,577	\$3,721	\$2,528	\$3,344	\$2,585	\$4,094	\$2,898
Buick Century	\$2,420	\$4,162	\$3,161	\$2,939	\$2,405	\$4,030	\$3,321
Nissan Maxima	\$2,881	\$4,674	\$4,565	\$2,085	\$2,911	\$6,941	\$3,513
Acura Legend	\$4,671	\$7,105	\$6,563	\$3,059	\$4,661	\$8,305	\$5,081
Lincoln Town Car	\$5,596	\$8,029	\$6,778	\$4,765	\$5,508	\$7,114	\$6,518
Cadillac Seville	\$7,500	\$10,733	\$8,635	\$4,863	\$7,439	\$9,182	\$8,015
Lexus LS400	\$9,030	\$10,510	\$8,411	\$4,791	\$8,585	\$10,925	\$7,398
BMW 735i	\$10,975	\$13,646	\$9,122	\$7,605	\$10,713	\$12,153	\$12,202
No. of demand side variables significant at 95% level ^a	5 of 5	4 of 5	4 of 5	6 of 6	4 of 5	8 of 8	4 of 6

^aA demand side variable is considered significant if either its mean or standard deviation (σ_p) is significant. See text for details.

- Estimated markups generally rise with price.
- The Mazda 323 has an estimated markup of about **\$801**; the Nissan Sentra about **\$880**; the Ford Escort about **\$1,077**; and the Chevy Cavalier about **\$1,302**.
- The Lexus LS400 has an estimated markup of about **\$9,030**; the BMW 735i about **\$10,975**.
- High markups do not always imply high total profits because sales volumes differ substantially.
- The Honda Accord and Ford Taurus have especially high estimated variable profits because they combine large volumes with sizable markups.

Economic message. BLP produces economically plausible markups: mass-market cars have lower per-unit markups, while differentiated luxury cars have high per-unit markups.

5.8 Table IX: alternative specifications

Read:

- The authors examine several robustness specifications, including adding $\log(q)$ to cost, replacing air conditioning with automatic transmission, splitting horsepower and weight, adding cost interactions, using region dummies and reliability, and adding weight with cost interactions.
- The general pattern of markups is robust across most specifications.
- The specification that separates horsepower and weight produces lower markups for expensive cars, which the authors view as less plausible.
- Adding cost interactions produces markup estimates very similar to the baseline.

Economic message. The exact magnitudes vary, but the main conclusion is stable: the random-coefficients demand and multiproduct pricing model generates sensible substitution and markup patterns.

6 Short summary

- The paper builds a structural model of differentiated-products demand and oligopoly pricing.
- Consumers choose among many automobile models and an outside option.
- Products differ in price, size, fuel economy, horsepower per weight, air conditioning, and unobserved quality.
- Consumer preferences are heterogeneous, so substitution depends on product similarity.
- Prices are endogenous because firms observe product quality that the econometrician does not.
- Instruments are built from observed characteristics of own and rival products.
- Firms set prices as multiproduct Bertrand-Nash competitors.
- Estimated demand elasticities and markups are much more plausible under BLP than under simple logit.
- The paper's inversion and contraction mapping make random-coefficients demand feasible with aggregate product-level data.
- The framework becomes the foundation for modern empirical industrial organization.

7 Conclusion

7.1 Core conclusion

Berry, Levinsohn, and Pakes show how to estimate a structural demand-and-supply model for differentiated products using aggregate product-level data. The method corrects price endogeneity, allows realistic substitution patterns, recovers marginal costs and markups, and can be used for policy counterfactuals.

7.2 Economic intuition

Consumers are not identical. Some care strongly about price, others about size, fuel economy, horsepower, or luxury. Firms understand this heterogeneity and set prices strategically across portfolios of products. Market shares and prices therefore jointly reveal information about tastes, marginal costs, and market power.

7.3 Takeaway

The central lesson is methodological and economic: to study markets with differentiated products, one needs a model that links individual choice, aggregate shares, price endogeneity, firm ownership, and equilibrium pricing. BLP provides that link and turns product-level data into estimates of demand elasticities, substitution patterns, marginal costs, and markups.